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THIS SPECIFICATION IS APPROVED FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE

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CONFIGURATION OF NUT OPTIONAL WITHIN THE LIMITATIONS IMPOSED BY DIMENSIONS AND REQUIREMENTS SPECIFIED HEREIN

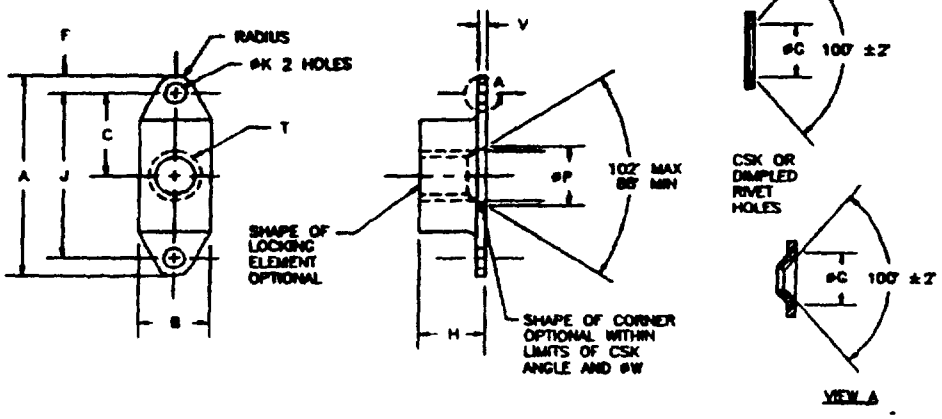


TABLE I DASH NUMBERS AND DIMENSIONS

DASH NO		T	A	B		C	F	øG	H		J	øK 1/	øP	
PLAIN HOLES	CSK OR DIMPLED HOLES	THREAD SIZE	MAX	MAX	MIN	± 005	MIN	± 010	MAX	MIN	± 010	+ 005 −.000	MAX	MIN
-04	-04K	1120-40UNJC-3B	984	416	328	344	115	200	218	140	688	098	260	198
-06	-06K	1380-32UNJC-3B	984	.416	.328	.344	115	.200	.234	.160	688	.098	.260	198
-08	-08K	1640-32UNJC-3B	984	416	.328	.344	115	.200	312	234	688	098	.270	.234
-3	-3K	.1900-32UNJF-3B	.984	.416	328	.344	115	.200	.312	.234	688	098	.270	.250
-4	-4K	.2500-28UNJF-3B	1.296	.516	406	.500	.115	.200	.387	283	1.000	098	.330	.310
-5	-5K	3125-24UNJF-3B	1.296	.609	484	.500	.115	.230	.387	283	1.000	130	.393	.373
-6	-6K	.3750-24UNJF-3B	1.296	641	563	500	115	.230	479	439	1 000	130	455	435
-7	---	.4375-20UNJF-3B	1.477	.719	672	.562	146	2/	.504	470	1 123	161	517	497
-8	---	5000-20UNJF-3B	1 802	.859	796	625	146	2/	630	596	1 250	161	580	560

TABLE I. Cont'd

DASH NO		V	øW		
PLAIN HOLES	CSK OR DIMPLED HOLES	MAX	MIN	AXIAL STRENGTH LB - MIN	WEIGHT LBS/100
-04	-04K	.075	112	750	.55
-06	-06K	.075	138	1 130	.55
-08	-08K	.075	164	1,720	.80
-3	-3K	.075	190	2,460	.80
-4	-4K	.075	260	4,880	1.20
-5	-5K	.075	.312	7,390	2.20
-6	-6K	075	375	11,450	2.80
-7	---	115	437	15,450	3 30
-8	---	115	500	21,110	4 00

1/ DIMPLED RIVET HOLE TOLERANCE FOR øK IS + 015, - 000
2/ NO COUNTERSUNK OR DIMPLED RIVET TYPE AVAILABLE FOR SIZES .4375 AND .5000.

(H) DENOTES CHANGE(S)

INCH-POUND

PREPARING ACTIVITY: DLA-IS
CUSTODIANS: ARMY- AV NAVY- AS AIR FORCE- 99
REVIEW: 82
USER: CR, MI
PROJECT NUMBER: 5310-1963

MILITARY SPECIFICATION SHEET

TITLE: NUT, SELF-LOCKING, PLATE, TWO LUG, FLOATING, NON-METALLIC INSERT, STEEL, 125 KSI Ftu, 250°F

SPECIFICATION SHEET NUMBER: MS21077 — 1 FEB 94 REV N

SUPERSEDING: MS21077 G — 16 DEC 91 (SEE NOTE 5.)

AMSC- N/A — FSC 5310

DISTRIBUTION STATEMENT
A. Approved for public release, distribution is unlimited

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REQUIREMENTS.

(H) 1 MATERIAL NUT- CARBON STEEL GRADE 1008 (UNS G10080) PER AMS 5081, 5082, GRADE 1035 (UNS G10350) PER AMS 5080, GRADE 1040 (UNS G10400) PER ASTM A827, GRADE 1042 (UNS G10420) PER ASTM A29, GRADE 1050 (UNS G10500) PER AMS 5085, GRADE 1137 (UNS G11370) PER AMS 5024, GRADE 11L37 (UNS G11374) PER AMS 5020 GRADE 1141 (UNS G11410) PER ASTM A29, GRADE 12L14 (UNS G12144) PER ASTM A108
ALLOY STEEL GRADE 4130 (UNS G4130) PER AMS 6370, GRADE 8740 (UNS G87400) PER MIL-S-6049

SHIM - (IF APPLICABLE)
CARBON STEEL GRADE 1008 (UNS G10080) PER AMS 5062, GRADE 1010 (UNS G10100) PER AMS 5040
ALUMINUM ALLOY 1100 ANNEALED (UNS A91100) PER QQ-A-250/1

RETAINER - (IF APPLICABLE)
CARBON STEEL GRADE 1050 (UNS G10080) PER AMS 5062, GRADE 1009 (UNS G10090) PER AMS 5062, GRADE 1010 (UNS G10100) PER AMS 5040, GRADE 1050 (UNS G10500) PER AMS 5085
ALUMINUM ALLOY 2024, TEMPER T4 (UNS A92024) PER QQ-A-250/4
ALUMINUM ALLOY 7075, TEMPER T6 (UNS A97075) PER QQ-A-250/12 OR QQ-A-250/13

INSERT - NYLON PER ASTM D 4066, GROUP 1, CLASS 1 OR 2

2 FINISH. STEEL - CADMIUM PLATE IN ACCORDANCE WITH QQ-P-416, TYPE II, CLASS 2
ALUMINUM - ANODIZE OR CHEMICALLY TREAT IN ACCORDANCE WITH PROCUREMENT SPECIFICATION

3 DIMENSIONING AND TOLERANCING. IN ACCORDANCE WITH ANSI/ASME Y14.5M

4 VIBRATION LIFE. MINIMUM VIBRATION LIFE REQUIREMENTS SHALL BE THREE TIMES AVERAGE VIBRATION LIFE IN ACCORDANCE WITH PROCUREMENT SPECIFICATION

5 THREADS. THREADS BEFORE LUBRICATION IN ACCORDANCE WITH MIL-S-8879

6 SURFACE TEXTURE. IN ACCORDANCE WITH ANSI/ASME B46.1 THE SURFACE TEXTURE SHALL NOT EXCEED 125 MICROINCHES.

7 LUBRICANT. DRY FILM LUBRICANT APPROVED IN ACCORDANCE WITH MIL-N-25027 OTHER LUBRICANTS SOLUBLE IN THE CLEANER SPECIFIED IN MIL-S-8802.

8 EDGES. BREAK SHARP EDGES AND REMOVE ALL BURRS.

9 FLOAT. FLOAT OF THE NUT PORTION OF ASSEMBLY SHALL NOT BE LESS THAN .030 LITERALLY AND LONGITUDINALLY FROM CENTERED POSITION. NUT BODY SHALL BE CAPABLE OF ENGAGEMENT WITH A BOLT IN MAXIMUM MISALIGNED POSITION. MAXIMUM AXIAL FLOAT .020 INCHES FOR 1900 AND SMALLER, .030 FOR .2500 AND LARGER. NUT MISALIGNMENT SHALL NOT EXCEED DIMENSION "B". THE NUT AND BASE PORTION OF THE ASSEMBLY SHALL FORM ONE INTEGRAL UNIT AND THE ASSEMBLY SHALL PROVIDE A BEARING SURFACE FOR THE NUT

10 EXAMPLE OF PART NUMBER: MS21077-4 = .2500-28 UNJF-3B NUT, SELF-LOCKING, PLATE, FLOATING, NON-METALLIC INSERT, STEEL, 125 KSI Ftu, 250 DEG. TWO LUG, PLAIN HOLES

- MS21077-4
  - NUT DASH NUMBER
  - BASIC MS PART NUMBER

NOTES.

1 DIMENSIONS ARE IN INCHES

2 IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.

3 REFERENCED GOVERNMENT (OR NON-GOVERNMENT) DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATION AND STANDARDS (DODISS) SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN.

4 DESIGN AND USAGE LIMITATIONS. THESE NUTS ARE DESIGNED TO DEVELOP THE TENSILE STRENGTH OF BOLTS AND SCREWS WITH AN ULTIMATE TENSILE STRENGTH OF 125 KSI BASED ON THE CROSS SECTIONAL AREA AT THE BASIC DIAMETER OF THE THREAD. THESE NUTS ARE DESIGNED TO BE USED ON 3A EXTERNALS THREADS. THESE NUTS SHALL BE USED IN ACCORDANCE WITH THE LIMITATIONS OF MS33588. ONLY NUTS FOR WHICH THERE ARE QUALIFIED PRODUCTS LISTED IN QPL25027 SHALL BE USED

5 MS21077 SUPERSEDES NAS1031

PREPARING ACTIVITY: DLA-IS CUSTODIANS: ARMY- AV NAVY- AS AIR FORCE- 99 REVIEW: AIR FORCE-82 USER: ARMY-CR, MI PROJECT NUMBER: 5310-1963	MILITARY SPECIFICATION SHEET TITLE NUT, SELF-LOCKING, PLATE, TWO LUG FLOATING, NON-METALLIC INSERT, STEEL, 125 KSI Ftu, 250°F	SPECIFICATION SHEET NUMBER MS21077 1 FEB 94 REV H SUPERSEDING MS21077 G 16 DEC 91 (SEE NOTE 5.) AMSC- N/A FSC 5310
DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.		Page 2 of 3

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INTERCHANGEABILITY RELATIONSHIP

MS21077 NUTS CAN UNIVERSALLY REPLACE NAS1031 NUTS OF LIKE MATERIAL, THREAD SIZE AND FASTENING METHOD (PLAIN RIVET HOLES OR COUNTERSUNK RIVET HOLES) BUT THESE NAS1031 NUTS CANNOT UNIVERSALLY REPLACE MS21077 NUTS.

TABLE II. INTERCHANGEABILITY.

CANCELLED PART NUMBERS	SUBSTITUTIVE PART NUMBERS
NAS1031N04	MS21077-04
NAS1031N04K	MS21077-04K
NAS1031N06	MS21077-06
NAS1031N06K	MS21077-06K
NAS1031N08	MS21077-08
NAS1031N08K	MS21077-08K
NAS1031N3	MS21077-3
NAS1031N3K	MS21077-3K
NAS1031N4	MS21077-04
NAS1031N4K	MS21077-4K
NAS1031N5	MS21077-5
NAS1031N5K	MS21077-5K
NAS1031N6	MS21077-6
NAS1031N7	MS21077-7
NAS1031N8	MS21077-8

PREPARING ACTIVITY: DLA-IS CUSTODIANS: ARMY- AV NAVY- AS AIR FORCE- 99 REVIEW: 82 USER: CR, MI PROJECT NUMBER: 5310-1983	MILITARY SPECIFICATION SHEET TITLE NUT, SELF-LOCKING, PLATE, TWO LUG, FLOATING, NON-METALLIC INSERT, STEEL, 125 KSI Ftu, 250°F	SPECIFICATION SHEET NUMBER MS21077 1 FEB 94 REV H SUPERSEDING MS21077 G 16 DEC 81 (SEE NOTE 5) AMSC- N/A FSC 5310
DISTRIBUTION STATEMENT A. Approved for public release distribution is unlimited		Page 3 of 3

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